

Yi-Hsuan Yang (楊奕軒)^{1,2}

1 Professor

Department of Electrical Engineering
National Taiwan University

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EXPERTISE

Music information research; Artificial intelligence; Machine learning; Music generation; Affective Computing

EDUCATION

- Ph.D., Communication Engineering, National Taiwan University, Taiwan 2010
- B.S., Electrical Engineering, National Taiwan University, Taiwan 2006

WORK EXPERIENCES

- **Full Professor**, Dept. Electrical Engineering, National Taiwan University **since 2023/02**
- Joint-Appointment Researcher, Research Center for IT Innovation, Academia Sinica **since 2023/08**
- **Chief Music Scientist**, Taiwan AI Labs **2019-2023**
- **Associate Research Professor**, Research Center for IT Innovation, Academia Sinica **2015-2023**
- Joint-Appointment Associate Professor, CSIE, National Cheng Kung University 2017-2019
- Adjunct Associate Professor, CSIE, National Tsing-Hua University 2016
- **Assistant Research Professor**, Research Center for IT Innovation, Academia Sinica **2011-2015**
- Visiting Scholar (three months), Columbia University, USA 2013
- Visiting Scholar (three months), Music Technology Group, Universitat Pompeu Fabra, Spain 2011
- Second Lieutenant (one year), Communications, Electronics and Information, ROC Army 2010-2011

AWARDS & HONORS

- Outstanding Teaching Awards, National Taiwan University 2025
- NTU-Fubon Distinguished Chair Professorship Award 2023
- Multimedia Rising Stars Award, IEEE International Conference on Multimedia Expo. (ICME) 2019
- Best Associate Editor Service Award, IEEE Transactions on Multimedia 2018
- Best Conference Paper Award, IEEE Multimedia Communications Technical Committee (MMTC) 2015
- Best Paper Award, IEEE International Conference on Multimedia Expo. (ICME) 2015

- Young Scholars' Creativity Award, Foundations for the Advancement of Outstanding Scholarship 2015
- Ta-You Wu Memorial Research Award, Ministry of Science and Technology 2014
- Best Poster Award, IEEE/ACM Joint Conference on Digital Libraries 2014
- Project for Excellent Junior Research Investigators, National Science Council 2013–2016
- Career Development Award, Academia Sinica 2013–2017
- Pan Wen Yuan Research Exploration Award 2013
- First Prize, ACM Multimedia Grand Challenge 2012
- IEEE SPS Young Author Best Paper Award, IEEE Signal Processing Society 2011
- Best Ph.D. Dissertation Award, Graduate Institute of Communication Engineering, NTU 2010
- Best Ph.D. Dissertation Award, TAAI (Taiwanese Association for Artificial Intelligence) 2010
- MediaTek Fellowship 2009
- Microsoft Research Asia (MSRA) Fellowship 2008

SELECTED RECENT PUBLICATIONS

- "MuseControlLite: Multifunctional music generation with lightweight conditioners," *ICML* 2025
- "MuseMorphose: Full-song and fine-grained piano music style transfer with just one Transformer VAE," *TASLP* 2023.
- "Relative positional encoding for Transformers with linear complexity," *ICML* 2021.
- "Compound Word Transformer: Learning to compose full-song music over dynamic directed hypergraphs," *AAAI* 2021.
- "Pop Music Transformer: Beat-based modeling and generation of expressive Pop piano compositions," *ACM Multimedia* 2020.
- "Musical composition style transfer via disentangled timbre representations," *IJCAI* 2019.
- "Score-to-audio music generation with multi-band convolutional residual network," *AAAI* 2019.
- "MuseGAN: Multi-track sequential GAN for symbolic music generation and accompaniment," *AAAI* 2018.
- "Generating music medleys via playing music puzzle games," *AAAI* 2018
- "MidiNet: A convolutional GAN for symbolic-domain music generation," *ISMIR* 2017
- *Music Emotion Recognition*, CRC Taylor & Francis Books, Feb. 2011.

ACADEMIC SERVICES

- **Test-of-Time Award Committee Member of**
Int. Society for Music Information Retrieval Conference (ISMIR) 2025
- **Associate Editor of**
IEEE Transactions on Multimedia 2016/9-2019/2
IEEE Transactions on Affective Computing 2016/11-2019/2
- **IEEE Senior Member** since 2017
- **Program Chair of**
Int. Society for Music Information Retrieval Conference (ISMIR) 2014

- **Guest Editor of**
 - ACM Transactions on Intelligent Systems and Technology 2015
 - IEEE Transactions on Affective Computing 2014
- **10K Award Committee Member of**
 - IEEE International Conference on Multimedia and Expo. (ICME) 2016–2018
- **Tutorial Chair of**
 - Int. Society for Music Information Retrieval Conference (ISMIR) 2021
- **Unconference Chair of**
 - Int. Society for Music Information Retrieval Conference (ISMIR) 2017
- **External PhD thesis committee member of**
 - Hong Kong University of Science and Technology 2015
- **Senior PC Member (Meta-reviewer) of**
 - AAAI 2022, ISMIR 2021, etc
- **Organizer of**
 - ICME 2026 Grand Challenge on Academic Text-to-Music Generation 2026
 - Int. Workshop on Affect and Sentiment in Multimedia, in conjunction with ACM MM 2015
 - MediaEval Affect Task: Music in Emotion 2013–2015
 - MIREX Singing Voice Separation Task 2014–2015
 - Int. Workshop on Affective Analysis in Multimedia, in conjunction with IEEE ICME 2013
 - Taiwanese Workshop on Music Information Retrieval 2012–2014

PROJECTS

- Towards Controllable and Affordable Text-to-Music Generation MOST 2025-2028
- Lightweight and Controllable Long-form Music Generation Google Gift 2025
- Controllable and Playable Music Generation Google Gift 2024
- Human-level and Open-Source Mandarin Pop Music Melody and Singing Generation by AI MOST 2023-2025
- Open DJ Project (II): Automatic EDM Generation MOST 2020-2022
- GenMusic Project: Industrial AI-Powered Music Composition Platform MOST 2018-2020
- Open DJ Project: AI for Automatic and Personalized DJing MOST 2018-2020
- A Unified Framework for Processing and Understanding Heterogeneous Data for Intelligent Recommendation (co-PI) MOST 2017-2020
- Product Recommendation and Customer Status Prediction Cathay 2017-2018
- Online Guitar Transcription: Melody, Chord and Playing Techniques Recognition MOST 2016-2018
- Mobile Music Recommendation using Brain-Computer Interfaces MOST 2015-2018
- User-centered Intelligent Music Streaming and Recommendation Platform (III) KKBOX Inc., 2017-2019
- User-centered Intelligent Music Streaming and Recommendation Platform (II) KKBOX Inc., 2015-2017

- User-centered Intelligent Music Streaming and Recommendation Platform KKBOX Inc., 2013-2015
- User Preference Modeling from Listening History & Artist Similarity KKBOX Inc., 2012-2013
- Music Recommendation based on Listening Context HTC Inc., 2012
- Dictionary-based Music Signal Analysis, Understanding, and Retrieval Academia Sinica, 2013-2017
- Automatic Music Recommendation and Retrieval MOST 2013-2016
- Dictionary-based Multipitch Estimation of Polyphonic Music NSC 2012-2013
- Large-scale Music Emotion Recognition System using Social Media NSC 2011-2012

TUTORIALS

- Hao-Wen Dong and **Yi-Hsuan Yang**, “Generating Music with GANs: An Overview and Case Studies,” *Int. Society for Music Information Retrieval Conference (ISMIR)*, 2019 ([link](#)).
- Xiao Hu and **Yi-Hsuan Yang**, “Music Affect Recognition: The State-of-the-art and Lessons Learned,” *Int. Society for Music Information Retrieval Conference (ISMIR)*, 2012.

STUDENT AWARDS

- 謝翔、王竣平、張譯心, 3rd Prize, Bachelor Thesis Award, Dept. of EE, NTU 2025
- Yi-Jen Shih, 1st Prize, Bachelor Thesis Award, Dept. of EE, NTU 2022
- Shih-Lun Wu, Ssu-Nien Fu's Award (1st Prize), Best Bachelor's Thesis, National Taiwan University 2021
- Shih-Lun Wu, 1st Prize, Bachelor Thesis Award, Dept. of CSIE, NTU 2021
- Yi-Hui Chou, 2nd Prize, Bachelor Thesis Award, Dept. of EE, NTU 2021
- Shih-Lun Wu, 1st Prize, Bachelor Thesis Award, Dept. of CSIE, NTU 2020
- Ching-Yu Chiu, Jury's Recommendation Award, 美律電聲論文獎 2021
- Yu-Hsiang Huang, Special Award, 美律電聲論文獎 2020
- Wen-Yi Hsiao, 2nd Prize, 美律電聲論文獎 2018

PUBLICATIONS

Total citations: 13,408; citations of most-cited paper: 981; h-index: 57; i10-index: 192

- **Book**

- [1] Y.-H. Yang and H. H. Chen, *Music Emotion Recognition*, CRC Taylor & Francis Books, Feb. 2011.

- **Proceedings (Edited)**

- [2] Meinard Müller, Emilia Gómez, and Yi-Hsuan Yang, “Computational methods for melody and voice processing in music recordings,” Report from Dagstuhl Seminar 19052, 2019.
- [3] Hsin-Min Wang, Yi-Hsuan Yang, and Jin Ha Lee, International Society for Music Information Retrieval Conference, Proceedings, ISMIR, Taipei, Taiwan, 2014.

- **Journal Papers**

- [4] Xavier Serra et al, "AI and music at a crossroads: The role of the MIR community in shaping trustworthy, open, and culturally inclusive Music-AI," *Transactions of the International Society for Music Information Retrieval*, 2026.
- [5] Gaël Richard, Vincent Lostanlen, Yi-Hsuan Yang, and Meinard Müller, "Model-based deep learning for music information research," *IEEE Signal Processing Magazine*, vol. 41, no. 6, pp. 51-59, November 2024
- [6] Yi-Hui Chou, I-Chun Chen, Chin-Jui Chang, Joann Ching, and Yi-Hsuan Yang, "MidiBERT-Piano: BERT-like pre-training for symbolic piano music classification tasks," *Journal of Creative Music Systems (JCMS)*, vol. 8, no. 1, 2024.
- [7] Ching-Yu Chiu, Meinard Müller, Matthew E. P. Davies, Alvin Wen-Yu Su, and Yi-Hsuan Yang, "Local periodicity-based beat tracking for expressive classical piano music," *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 31, pp. 2824-2835, July 2023.
- [8] Shih-Lun Wu and Yi-Hsuan Yang, "MuseMorphose: Full-song and fine-grained piano music style transfer with just one Transformer VAE," *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 31, pp. 1953-1967, May 2023.
- [9] Ching-Yu Chiu, Meinard Müller, Matthew E. P. Davies, Alvin Wen-Yu Su, and Yi-Hsuan Yang, "An analysis method for metric-level switching in beat tracking," *IEEE Signal Processing Letters (SPL)*, vol. 29, pp. 2153-2157, Oct. 2022.
- [10] Yi-Jen Shih, Shih-Lun Wu, Frank Zalkow, Meinard Müller, and Yi-Hsuan Yang, "Theme Transformer: Symbolic music generation with theme-conditioned Transformer," *IEEE Transactions on Multimedia (TMM)*, vol. 25, pp. 3495-3508, March 2022.
- [11] Juan Sebastián Gomez-Cañón, Estefanía Cano, Tuomas Eerola, Perfecto Herrera, Xiao Hu, Yi-Hsuan Yang, and Emilia Gómez, "Music Emotion Recognition: Towards new robust standards in personalized and context-sensitive applications," *IEEE Signal Processing Magazine*, vol. 38, no. 6, pp. 106-114, Nov. 2021.
- [12] Ching-Yu Chiu, Alvin Wen-Yu Su, and Yi-Hsuan Yang, "Drum-aware ensemble architecture for improved joint musical beat and downbeat tracking," *IEEE Signal Processing Letters (SPL)*, vol. 28, pp. 1100-1104, May 2021.
- [13] Yin-Cheng Yeh, Wen-Yi Hsiao, Satoru Fukayama, Tetsuro Kitahara, Benjamin Genchel, Hao-Min Liu, Hao-Wen Dong, Yian Chen, Terence Leong, and Yi-Hsuan Yang, "Automatic melody harmonization with triad chords: A comparative study," *Journal of New Music Research*, vol. 50, no. 1, pp. 37-51, 2021.
- [14] E. Zangerle, C.-M. Chen, M.-F. Tsai and Y.-H. Yang, "Leveraging affective hashtags for ranking music recommendations," *IEEE Transactions on Affective Computing (TAC)*, vol. 12, no. 1, pp. 78-91, 2021.
- [15] Zhe-Cheng Fan, Tak-Shing T. Chan, Yi-Hsuan Yang, and Jyh-Shing R. Jang, "Backpropagation with N -D vector-valued neurons using arbitrary bilinear products," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 31, no. 7, pp. 2638-2652, 2020.
- [16] T.-W. Su, Y.-P. Chen, L. Su, and Y.-H. Yang, "TENT: Technique-embedded note tracking for real-world guitar solo recordings," *Transactions of the International Society for Music Information Retrieval (TISMIR)*, vol. 2, no. 1, pp. 15-28, 2019.
- [17] S.-Y. Chou, J.-S. R. Jang, and Y.-H. Yang, "Fast tensor factorization for large-scale context-aware recommendation from implicit feedback," *IEEE Trans. Big Data (TBD)*, vol. 6, no. 1, pp. 201-208, Mar. 2020.

- [18] J.-Y. Liu, Y.-H. Yang, and S.-K. Jeng, "Weakly-supervised visual instrument-playing action detection in videos," *IEEE Transactions on Multimedia* (TMM), vol. 21, no. 4, pp. 887-901, Apr. 2019.
- [19] J. Nam, K. Choi, J. Lee, S.-Y. Chou, and Y.-H. Yang, "Deep learning for audio-based music classification and tagging," *IEEE Signal Processing Magazine* (SPM), vol. 36, no. 1, pp. 41-51, Jan. 2019.
- [20] J.-C. Lin, W.-L. Wei, T.-L. Liu, Y.-H. Yang, H.-M. Wang, H.-R. Tyan, and H.-Y. M. Liao, "Coherent deep-net fusion to classify shots in concert videos," *IEEE Transactions on Multimedia* (TMM), vol. 20, no. 11, pp. 3123-3136, Nov. 2018.
- [21] Y.-H. Chin, J.-C. Wang, J.-C. Wang and Y.-H. Yang, "Predicting the probability density function of music emotion using emotion space mapping," *IEEE Transactions on Affective Computing* (TAC), vol. 9, no. 4, pp. 541-549, Oct.-Dec. 2018.
- [22] Y.-S. Huang, S.-Y. Chou, and Y.-H. Yang, "Pop music highlighter: Marking the emotion keypoints," *Transactions of the International Society for Music Information Retrieval* (TISMIR), vol. 1, no. 1, pp. 68-78, Sep. 2018.
- [23] Y.-P. Lin, P.-K. Jao, and Y.-H. Yang, "Improving cross-day EEG-based emotion classification using robust principal component analysis," *Frontiers in Computational Neuroscience*, Jul. 2017.
- [24] A. Aljanaki, Y.-H. Yang, and M. Soleymani, "Developing a benchmark for emotional analysis of music," *PLOS ONE*, vol. 12, no. 3, e0173392.doi:10.1371/journal.pone.0173392, Mar. 2017.
- [25] X. Hu and Y.-H. Yang, "The mood of Chinese pop music: Representation and recognition," *Journal of the Association for Information Science and Technology* (JAIST), doi:10.1002/asi.23813, Jun. 2017.
- [26] Y.-A. Chen, J.-C. Wang, Y.-H. Yang, H. H. Chen, "Component tying for mixture model adaptation in personalization of music emotion recognition," *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (TASLP), vol. 25, no. 7, pp. 1409-1420, Jul. 2017. [cover page of the issue]
- [27] X. Hu and Y.-H. Yang, "Cross-dataset and cross-cultural music mood prediction: A case on Western and Chinese pop songs," *IEEE Transactions on Affective Computing* (TAC), vol. 8, no. 2, pp. 228-240, Apr. 2017.
- [28] T.-S. Chan and Y.-H. Yang, "Informed group-sparse representation for singing voice separation," *IEEE Signal Processing Letters* (SPL), vol. 24, no. 2, pp. 156-160, Feb. 2017.
- [29] T.-S. Chan and Y.-H. Yang, "Polar n -complex and n -bicomplex singular value decomposition and principal component pursuit," *IEEE Transactions on Signal Processing* (TSP), vol. 64, no. 24, pp. 6533-6544, Dec. 2016.
- [30] M. Schedl, Y.-H. Yang, and P. Herrera, "Introduction to intelligent music systems and applications," *ACM Transactions on Intelligent Systems and Technology* (TIST), vol. 8, no. 2, article 17, Oct. 2016.
- [31] P.-K. Jao, L. Su, Y.-H. Yang and B. Wohlberg, "Monaural music source separation using convolutional sparse coding," *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (TASLP), vol. 24, no. 11, pp. 2158-2170, Nov. 2016.
- [32] T.-S. Chan and Y.-H. Yang, "Complex and quaternionic principal component pursuit and its application to audio separation," *IEEE Signal Processing Letters* (SPL), vol. 23, no. 2, pp. 287-291, Feb. 2016.
- [33] C.-Y. Liang, L. Su and Y.-H. Yang, "Musical onset detection using constrained linear reconstruction," *IEEE Signal Processing Letters* (SPL), vol. 22, no. 11, pp. 2142-2146, Nov. 2015.
- [34] L. Su and Y.-H. Yang, "Combining spectral and temporal representations for multipitch estimation of

polyphonic music,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 23, no. 10, pp. 1600-1612, Oct. 2015.

- [35] P.-K. Jao and Y.-H. Yang, “Music annotation and retrieval using unlabeled exemplars: correlation and sparse codes,” *IEEE Signal Processing Letters (SPL)*, vol. 22, no. 10, pp. 1771-1775, Oct. 2015.
- [36] Y.-H. Yang and Y.-C. Teng, “Quantitative study of music listening behavior in a smartphone context,” *ACM Transactions on Interactive Intelligent Systems (TiiS)*, vol. 5, no. 3, article 14, Aug. 2015.
- [37] M. Soleymani, Y.-H. Yang, G. Irie, and A. Hanjalic, “Challenges and perspectives for affective analysis in multimedia,” *IEEE Transactions on Affective Computing (TAC)*, vol. 6, no. 3, pp. 206-208, 2015.
- [38] J.-C. Wang, Y.-H. Yang, H.-M. Wang, and S.-K. Jeng, “Modeling the affective content of music with a Gaussian mixture model,” *IEEE Transactions on Affective Computing (TAC)*, vol. 6, no. 1, pp. 56-68, Feb. 2015.
- [39] L. Su, H.-M. Lin, and Y.-H. Yang, “Sparse modeling of magnitude and phase-derived spectra for playing technique classification,” *IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 22, no. 12, pp. 2122-2132, Dec. 2014.
- [40] L. Su, C.-C. Yeh, J.-Y. Liu, J.-C. Wang, and Y.-H. Yang, “A systematic evaluation of the bag-of-frames representation for music information retrieval,” *IEEE Transactions on Multimedia (TMM)*, vol. 16, no. 5, pp. 1188-1200, Aug. 2014.
- [41] Y.-P. Lin, Y.-H. Yang, and T.-P. Jung, “Fusion of Electroencephalogram dynamics and musical contents for estimating emotional responses in music listening,” *Frontiers in Neuroscience*, vol. 8, no. 94, pp. 1-14, May 2014.
- [42] Y.-H. Yang and J.-Y. Liu, “Quantitative study of music listening behavior in a social and affective context,” *IEEE Transactions on Multimedia (TMM)*, vol. 15, no. 6, pp. 1304-1315, Oct. 2013.
- [43] K.-S. Lin, A. Lee, Y.-H. Yang, C.-T. Lee, and H. H. Chen, “Automatic highlights extraction for drama video using music emotion and human face features,” *Neurocomputing*, vol. 119, pp. 111–117, Nov. 2013.
- [44] C.-T. Lee, Y.-H. Yang and H. H. Chen, “Multipitch estimation of piano music by exemplar-based sparse representation,” *IEEE Transactions on Multimedia (TMM)*, vol. 14, no. 3, pp. 608–618, Jun. 2012.
- [45] Y.-H. Yang and H. H. Chen, “Machine recognition of music emotion: a review,” *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 3, no. 3, article 40, May 2012.
- [46] Y.-C. Lin, Y.-H. Yang, and H. H. Chen, “Exploiting online tags for music emotion classification,” *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMCCAP)*, vol. 7s, no. 1, article 26, Oct. 2011.
- [47] Y.-H. Yang and H. H. Chen, “Prediction of the distribution of perceived music emotions using discrete samples,” *IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 19, no. 7, pp. 2184 -2196, Sep. 2011.
- [48] Y.-H. Yang and H. H. Chen, “Ranking-based emotion recognition for music organization and retrieval,” *IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 19, no. 4, pp. 762-774, May 2011.
- [49] Y.-F. Su, Y.-H. Yang, M.-T. Lu, and H. H. Chen, “Smooth control of adaptive media playout for video streaming,” *IEEE Transactions on Multimedia (TMM)*, vol. 11, no. 7, pp. 1331–1339, Nov. 2009.
- [50] Y.-H. Yang, W.-H. Hsu, and H. H. Chen, “Online reranking via ordinal informative concepts for context

fusion in concept detection and video search,” *IEEE Transactions on Circuits and Systems for Video Technology* (TCSVT), vol. 19, no. 12, pp. 1880–1890, Dec. 2009.

- [51] Y.-H. Yang, Y.-C. Lin, Y.-F. Su, and H. H. Chen, “A regression approach to music emotion recognition,” *IEEE Transactions on Audio, Speech, and Language Processing* (TASLP), vol. 16, no. 2, pp. 448–457, Feb. 2008.

• Conference Papers

- [52] Chih-Heng Chang, Keng-Seng Ho, Chih-Yu Tsai, Kuan-Lin Chen, Yi-Hsuan Yang, and Jian-Jiun Ding, “AnchorSteer: Self-discovered concept injection for structure-preserving music editing,” in *Proc. ACM Conf. Knowledge Discovery and Data Mining* (KDD), 2026.
- [53] Fang-Chih Hsieh, Wei-Jaw Lee, Chun-Ping Wang, Hung-yi Lee, Hao-Wen Dong, Yi-Hsuan Yang, “Academic text-to-music grand challenge: Datasets, baselines, and evaluation methods,” *IEEE Int. Conf. Multimedia and Expo. (ICME)*, Grand Challenge Paper, 2026.
- [54] Jeng-Yue Liu, Ting-Chao Hsu, Yen-Tung Yeh, Li Su, and Yi-Hsuan Yang, “SynthCloner: Synthesizer preset conversion via factorized codec with ADSR envelope control,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing* (ICASSP), 2026.
- [55] Ting-Kang Wang, Chih-Pin Tan, and Yi-Hsuan Yang, “Time-shifted token scheduling for symbolic music generation,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing* (ICASSP), 2026.
- [56] Xuanjun Chen, Chia-Yu Hu, I-Ming Lin, Yi-Cheng Lin, I-Hsiang Chiu, You Zhang, Sung-Feng Huang, Yi-Hsuan Yang, Haibin Wu, and Jyh-Shing Roger Jang, “How does instrumental music help SingFake detection?,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing* (ICASSP), 2026.
- [57] Ting-Chao Hsu and Yi-Hsuan Yang, “Conditional vocal timbral technique conversion via embedding-guided dual attribute modulation,” in *AAAI Workshop on Artificial Emerging AI Technologies for Music* (EAIM), 2026.
- [58] Jia-Wei Liao, Pin-Chi Pan, Li-Xuan Peng, Sheng-Ping Yang, Yen-Tung Yeh, Cheng-Fu Chou, Yi-Hsuan Yang, “Zero-shot geometry-aware diffusion guidance for music restoration,” in *NeurIPS Workshop on Artificial Intelligence for Music* (AI4Music), 2025.
- [59] Ping-Yi Chen, Chih-Pin Tan, Yi-Hsuan Yang, “Segment-factorized full-song generation on symbolic piano music,” in *NeurIPS Workshop on Artificial Intelligence for Music* (AI4Music), 2025.
- [60] Hsin Ai and Yi-Hsuan Yang, “Transformer-based unpaired piano accompaniment style transfer,” in *Proc. Asia Pacific Signal and Information Processing Association Annual Summit and Conf. (APSIPA ASC)*, 2025.
- [61] Yen-Tung Yeh, Junghyun Koo, Marco Martínez-Ramírez, Wei-Hsiang Liao, Yi-Hsuan Yang, and Yuki Mitsufuji, “Fx-Encoder++: Extracting instrument-wise audio effect representations from mixtures,” in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2025.
- [62] Fang-Duo Tsai, Shih-Lun Wu, Weijaw Lee, Sheng-Ping Yang, Bo-Rui Chen, Hao-Chung Cheng, and Yi-Hsuan Yang, “MuseControlLite: Multifunctional music generation with lightweight conditioners,” in *Proc. Int. Conf. Machine Learning* (ICML), 2025.
- [63] Dinh-Viet-Toan Le and Yi-Hsuan Yang, “METEOR: Melody-aware texture-controllable symbolic music re-orchestration via Transformer VAE,” in *Proc. Int. Joint Conf. Artificial Intelligence* (IJCAI), 2025.
- [64] Yen-Tung Yeh, Yu-Hua Chen, Yuan-Chiao Cheng, Jui-Te Wu, Jun-Jie Fu, Yi-Fan Yeh, and Yi-Hsuan Yang, “DDSP Guitar Amp: Interpretable guitar amplifier modeling,” in *Proc. IEEE Int. Conf. Acoustics,*

Speech and Signal Processing (ICASSP), 2025.

- [65] Chon In Leong, I-Ling Chung, Kin Fong Chao, Jun-You Wang, Yi-Hsuan Yang, and Jyh-Shing Roger Jan, "Music2Fail: Transfer music to failed recorder style," in *Proc. Asia Pacific Signal and Information Processing Association Annual Summit and Conf. (APSIPAASC)*, 2024.
- [66] Jingyue Huang, Ke Chen, and Yi-Hsuan Yang, "Emotion-driven piano music generation via two-stage disentanglement and functional representation," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2024.
- [67] Yun-Han Lan, Wen-Yi Hsiao, Hao-Chung Cheng, Yi-Hsuan Yang, "MusiConGen: Rhythm and chord control for Transformer-based text-to-music generation," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2024.
- [68] Fang-Duo Tsai, Shih-Lun Wu, Haven Kim, Bo-Yu Chen, Hao-Chung Cheng, and Yi-Hsuan Yang, "Audio Prompt Adapter: Unleashing music editing abilities for text-to-music with lightweight finetuning," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2024.
- [69] Chih-Pin Tan, Hsin Ai, Yi-Hsin Chang, Shuen-Huei Guan, and Yi-Hsuan Yang, "PiCoGen2: Piano cover generation with transfer learning approach and weakly aligned data," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2024.
- [70] Yu-Hua Chen, Yen-Tung Yeh, Yuan-Chiao Cheng, Jui-Te Wu, Yu-Hsiang Ho, Jyh-Shing Roger Jang, and Yi-Hsuan Yang, "Towards zero-shot amplifier modeling: One-to-many amplifier modeling via tone embedding control," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2024.
- [71] Yen-Tung Yeh, Wen-Yi Hsiao and Yi-Hsuan Yang, "Hyper recurrent neural network: Condition mechanisms for black-box audio effect modeling," in *Proc. Int. Conf. Digital Audio Effects (DAFx)*, 2024.
- [72] Yu-Hua Chen, Woosung Choi, Wei-Hsiang Liao, Marco Martínez-Ramírez, Kin Wai Cheuk, Yuki Mitsufuji, Jyh-Shing Roger Jang and Yi-Hsuan Yang, "Improving unsupervised clean-to-rendered guitar tone transformation using GANs and integrated unaligned clean data," in *Proc. Int. Conf. Digital Audio Effects (DAFx)*, 2024.
- [73] Ying-Shuo Lee, Yueh-Po Peng, Jui-Te Wu, Ming Cheng, Li Su and Yi-Hsuan Yang, "Distortion recovery: A two-stage method for guitar effect removal," in *Proc. Int. Conf. Digital Audio Effects (DAFx)*, 2024.
- [74] Chih-Pin Tan, Shuen-Huei Guan, Yi-Hsuan Yang, "PiCoGen: Generate piano covers with a two-stage approach," in *Proc. ACM Int. Conf. Multimedia Retrieval (ICMR)*, 2024.
- [75] Shih-Lun Wu and Yi-Hsuan Yang, "Compose & Embellish: Well-structured piano performance generation via a two-stage approach," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
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